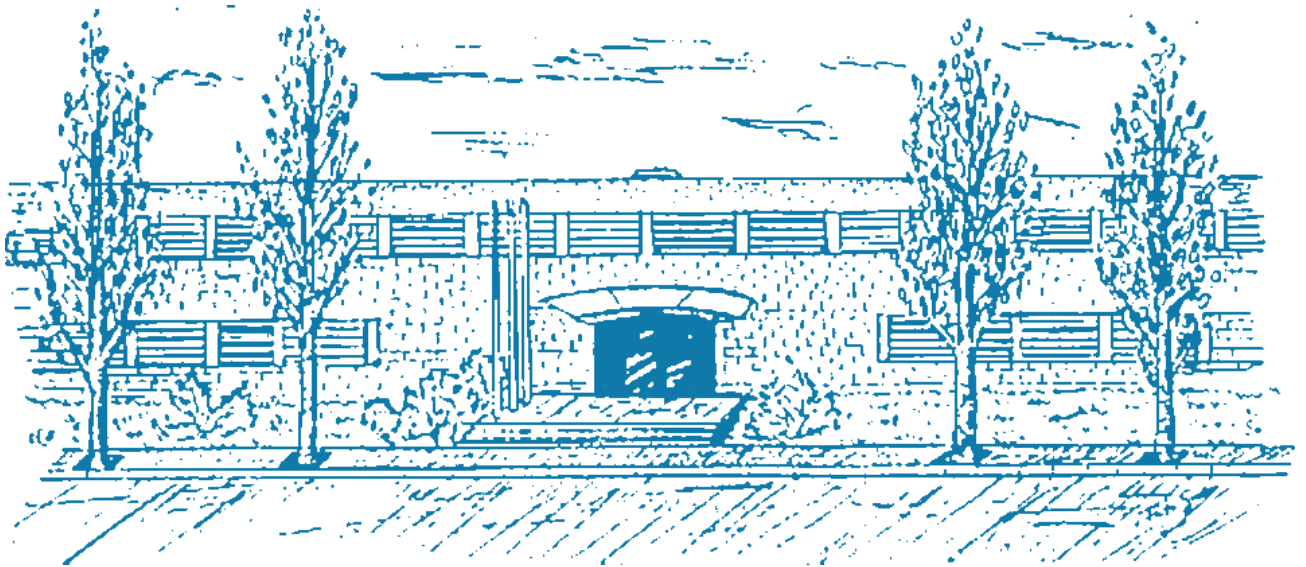


UNIVERSITAT POLITÈCNICA DE CATALUNYA - BarcelonaTech (UPC)
FME - FACULTAT DE MATEMÀTIQUES I ESTADÍSTICA

MASTER'S DEGREE IN STATISTICS AND OPERATIONS RESEARCH (MESIO UPC-UB)
MASTER'S THESIS

TRAFFIC SIMULATION WITH IRRATIONAL DRIVERS



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Abstract

Most traffic simulations assume drivers act rationally, but research in cognitive science shows that decisions are often biased and heuristic.

While traffic simulations often incorporate external factors such as weather conditions, fuel prices, and calendar effects, an important component is frequently simplified or overlooked: the individual decisions made by drivers. These decisions are commonly modeled as fully rational; however, decades of research in cognitive science demonstrate that human decision-making is systematically biased and frequently deviates from optimality.

This project bridges traffic simulation and cognitive modeling by first training rational route-choice agents with Reinforcement Learning in SUMO (Simulation of Urban MObility), an open-source microscopic traffic simulator, as the simulation platform for validation and testing. A second phase introduces cognitively inspired agents whose route choices emerge from a mixture of decision-making subsystems. While one subsystem will reflect the rational agent, others implement known human heuristics, often suboptimal strategies grounded in cognitive science. By parameterizing the influence of each subsystem on the final decision, we infer the distribution of strategies that best explain real-world traffic data.

This approach offers a novel framework for modeling urban mobility grounded in realistic human behavior, with potential applications in traffic forecasting, infrastructure planning, and behavioral intervention design.

Resumen

To God.

To my parents and my sisters, for their constant support.

To Mari Paz and Manuel Molano.

To those who encouraged me to pursue this master's degree.

And to David Goggins and other role models who helped me change my inner dialogue. As he says, "we are leaving so much on the table."

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1 Introduction

1.1 Background

1.2 Motivation and Thesis Objectives

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1.4 Thesis Outline

2 Literature Review

2.1 Overview

2.2 Shortest Path Algorithm

2.3 Dynamic Traffic Assignment

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3 Urban Traffic Simulation

In many cases, conducting experiments in real-world environments is impractical or resource-intensive. Simulation provides an effective alternative by enabling the study of systems within controlled artificial environments. One of its main advantages is the ability to evaluate system behavior prior to real-world deployment. With the continuous growth of computational power, simulation has become a practical tool for analyzing complex scientific models within reasonable time and cost constraints. These models rely on mathematical formulations to characterize system behavior. When a simulation accurately represents the underlying system, it can provide valuable insights for real-world applications, allowing controlled experimentation, time scaling, and cost reduction. This chapter presents an introduction to urban traffic simulation, along with an overview of SUMO, the traffic simulator used in this thesis.

3.1 Traffic Simulation

Traffic is one of the most commonly studied systems in simulation, as conducting real-world experiments with vehicles is often impractical. Simulation enables full control over system variables and allows scenarios to be run faster or slower than in reality—for example, compressing hours of traffic into minutes or slowing down events for detailed analysis—while reducing costs. However, traffic simulation is inherently complex, as it involves diverse interacting components, including vehicles and road networks, that model real-world transportation dynamics.

Traffic simulation models are commonly classified according to their level of granularity, ranging from low- to high-fidelity as follows:

- Macroscopic simulation models
- Mesoscopic simulation models
- Microscopic simulation models

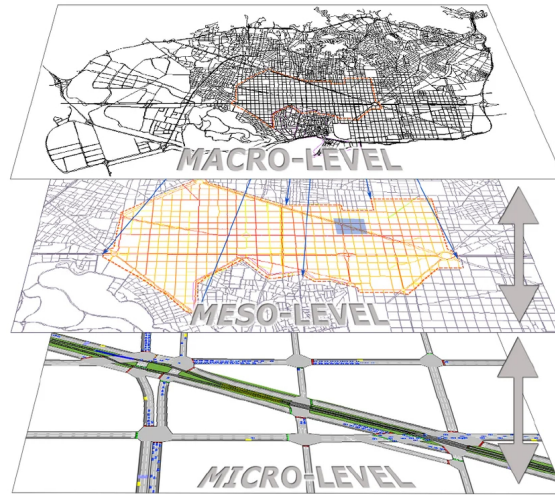


FIG. 1. Overview of traffic simulation models. Source: Barceló (2005)

Macroscopic models describe traffic as an aggregate flow, typically using formulations inspired by fluid dynamics, without explicitly representing individual vehicles. In this context, routing decisions are expressed in aggregate terms, for example by assigning proportions of traffic flow to different routes. In contrast, microscopic models capture the behavior and interactions of individual vehicles with each other and with the infrastructure at a high level of detail. They are often implemented in discrete time through simulation steps, following behavioral rules such as car-following and lane-changing models. Traveler decisions, such as route choice, are represented at the individual level, allowing each driver to select and adapt their route in response to traffic conditions. In addition, elements such as traffic control systems, infrastructure, and detectors are modeled in detail. Mesoscopic models provide an intermediate representation, combining aspects of both approaches by modeling individual vehicles with simplified interaction dynamics, thereby balancing realism and computational efficiency.

The thesis employs a microscopic simulator, as the objective is to analyze individual route choice behavior. Microscopic models can be further classified according to their spatial and temporal representations, which may be either discrete or continuous. A model is discrete in time if the system is updated at fixed time steps (e.g. every second), resulting in state transitions rather than continuous evolution. A model is discrete in space if the domain is divided into a set of finite cells, each

representing a portion of the road network and holding a specific state (e.g. empty or occupied).

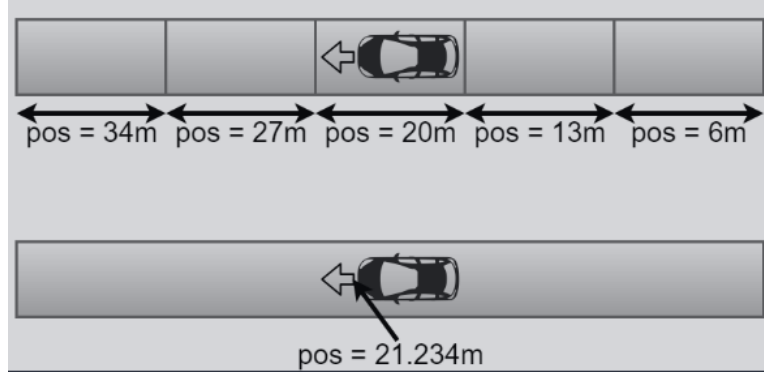


FIG. 2. Space-discrete vs Space-continuous simulation. Source: Koh (2022)

The microscopic simulator used in this thesis is SUMO (Simulation of Urban MObility). The main reasons for this choice are as follows:

- **Open-source:** SUMO is freely available and open-source.
- **Output capabilities:** SUMO provides a wide range of configuration options that enable the generation of detailed output files containing simulation data.
- **API:** SUMO provides the TraCI (Traffic Control Interface) API, which allows interaction with a running simulation. Through this interface, it is possible to retrieve information about simulated entities and modify their behavior in real time.
- **Performance and stability**

3.2 Overview of SUMO

The development of SUMO (Simulation of Urban MObility) was initiated in 2001 by the German Aerospace Center (DLR). It is a comprehensive suite of traffic simulation tools that covers a wide range of functionalities.

- **Network building:** SUMO provides multiple approaches for network creation. It includes tools for automatically generating synthetic networks, as well as importing real-world networks from external sources such as OpenStreetMap or other traffic simulators (e.g., VISUM) using the `netconvert` tool. Additionally, the `netedit` tool enables manual creation and modification of networks.
- **Demand modeling:** Demand in traffic simulation corresponds to the set of vehicles in the network.
 - (1) **Demand representation:** SUMO supports different demand definitions:

- **Trip:** Defines a single vehicle by specifying origin edge, destination edge and departure time. The route is computed automatically using `duarouter`
- **Flow:** Defines multiple vehicles over a time interval, specifying origin, destination and either a total number of vehicles or a flow rate.
- **Vehicle with predefined route:** Defines a vehicle with an explicitly specified route.

Unlike many traffic simulators that rely primarily on OD matrices, SUMO often represents demand using trips.

- (2) **Demand generation:** Demand can be generated from different data sources:
 - **OD matrices:** The `od2trips` tool converts OD matrices into trips.
 - **Traffic counts:** Tools such as `routeSampler` generate demand from turn counts or edge counts.
 - **Synthetic demand:** The `randomTrips` tool generates random trips when data is not available.
- (3) **Routing:** When demand is specified as trips, routes must be computed. SUMO provides several routing approaches:
 - Shortest path methods (e.g., Dijkstra or A*) via `duarouter`, typically based on free-flow travel times, and used as the default routing approach in SUMO.
 - Dynamic User Equilibrium (DUE) using `duaIterate`.
 - Additional approaches for achieving System Optimum (SO).
- (4) **Output:** SUMO can generate detailed output files describing different aspects of the simulation, such as trip information (`tripInfo`), vehicle routes (`vehroute`) among other outputs.
- (5) **API (TraCI):** SUMO provides the TraCI (Traffic Control Interface), which enables interaction with a running simulation. It allows both retrieval of simulation data (e.g., vehicle speed) and modification of system state (e.g., changing vehicle behavior) in real time.

In the day-to-day route choice learning algorithm (see subsection 5.1) , TraCI is not required, as the feedback signal received by the RL agents is obtained from output files generated after the simulation has finished. Therefore, no interaction with the simulation during execution is necessary.

In contrast, in the real-time vehicle route optimization algorithm (see subsection 5.2), TraCI is used as an interface between SUMO and the reinforcement learning (RL) agent. It enables the retrieval of relevant information about the environment (e.g., network state) to construct the agent’s state, as well as the application of the agent’s actions to the simulation in real time.
- (6) **Additional features:** SUMO includes many additional capabilities, such as traffic light control, multimodal simulation (e.g., pedestrians and bicycles), and vehicle emission modeling.

SUMO provides a high level of flexibility in modeling individual vehicles and their behavior. Traffic dynamics are governed by car-following and lane-changing models, which determine how vehicle accelerate, decelerate, and interact with their surroundings. These models take as input both vehicle-specific characteristics, such as length, maximum speed and acceleration capabilities, and parameters related to driver behavior, including impatience and cooperative behavior. In this work, the default parameter settings provided by SUMO are used to focus on the analysis of route choice behavior.

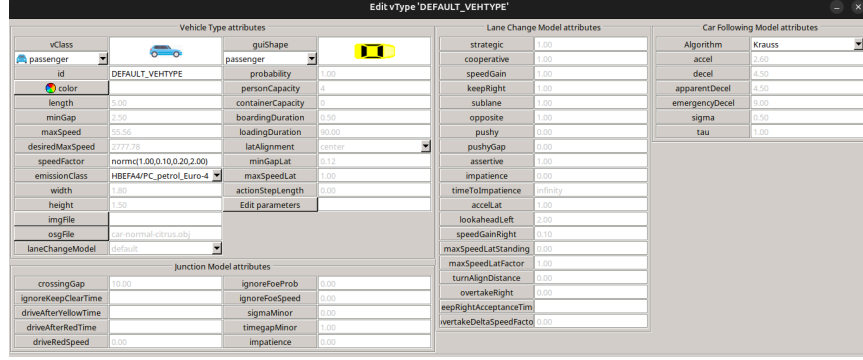


FIG. 3. Vehicle type configuration in SUMO, showing parameters related to car-following, lane-changing, and vehicle characteristics

In terms of the previously introduced taxonomy, SUMO can be classified as a microscopic traffic simulator, as it models individual vehicle behavior in detail. Additionally, it employs a continuous spatial representation and a discrete-time simulation framework, where vehicle states are updated at fixed simulation steps.

4 Reinforcement Learning

4.1 Bush-Mosteller

4.2 Deep Reinforcement Learning

5 Reproduction of Existing Methods

5.1 Day-to-Day Route Choice Model

5.1.1 Program Execution Modes

The program that implements this algorithm can be run in different modes.

- **compute_routes**: Computes k routes for each OD pair. This is the most computationally expensive step with large networks. The resulting routes are stored and can be reused in future executions.

- **train:** Main training procedure. It can either *compute routes internally*, or load a previously generated routes file. It allows selecting which episodes are visualized using the SUMO GUI.
- **eval_gui:** Post-processing mode used to visualize the last episode from a previous training run. It avoids re-running the full training process by reusing stored outputs.

The separation of *compute_routes* from *train* enables caching of expensive computations, significantly reducing execution time in repeated experiments.

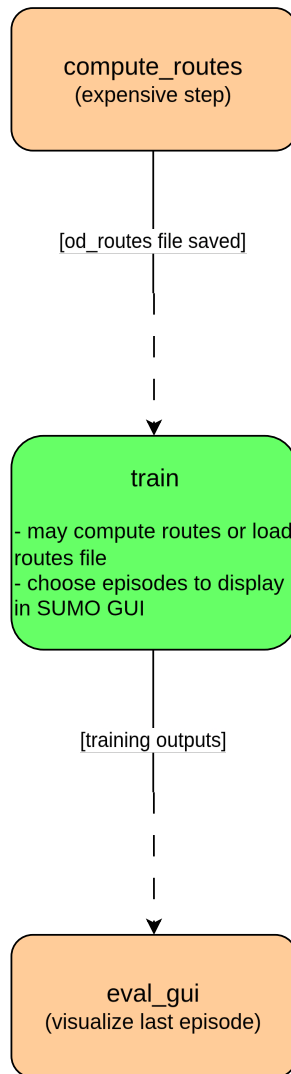


FIG. 4. Overview of program execution modes and their interactions

5.1.2 Experiment Evaluation**5.2 Realtime Vehicle Route Optimisation****5.2.1 Experiment Evaluation****6 Proposed Framework and Design****7 Experiment Implementation and Evaluation****8 Conclusion and Future Work****9 Bibliography****10 Anexos**